Received: 10 July 2022 | Revised: 21 November 2022 | Accepted: 5 January 2023

DOI: 10.1111/sms.14316

REVIEW

WILEY

Athletes are not at greater risk for death by suicide: A review

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Introduction: Suicide represents a major mental and public health issue. Elite athletes share certain individual and environmental characteristics that may increase their risk for mental illnesses, ultimately leading to suicide. This notion conflicts with the general perception of athletes, being the healthiest representatives of society.

Methods: A comprehensive literature search was carried out through PubMed and Embase databases for relevant publications.

Results: Recent calls for investigating suicidality among athletes resulted in a considerable amount of literature providing some evidence regarding lower rates of suicide among professional and high-performance athletes as well as similar incidence and prevalence of mental conditions, which are known as risk factors for suicide. Nevertheless, special attention is required in this population as predisposing and precipitating factors might differ from classical features of suicidality in the general population. Sports physicians, sports psychiatrists, and other mental health professionals in elite sports should be aware of early signs of affective disorders, risk of recreational drug abuse, misuse of performance-enhancing medications, sport-specific environmental stressors, serious physical injuries, and presence of physical or mental illness, all of which may increase suicidality. Traumatic brain injury (TBI) is with suicide with higher severity correlated with increased risk. Compared to active athletes, former athletes may have higher rates of suicide due to common life stressors occurring after sports retirement.

Conclusions: The findings suggest a multidisciplinary approach to suicidality in elite athletes, the main goal of which should be the reduction of suicide-related morbidity and mortality. Further research is required to clarify the existing gaps in the current knowledge of the issue. While having lower rates of suicide, athletes share some similar (affective disorders, drug abuse, mental and physical illness) and unique factors (misuse of performance-enhancing substances, sports-related stressors, sports injuries, TBI) putting them at risk of suicide during active career and retirement.

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KEYWORDS
athletes, professional sports, suicidal behavior, suicidal ideation, suicidality, suicide, suicide prevention, suicide risk

1 | INTRODUCTION

Suicide is a global public health concern being the tenth leading cause of death worldwide and claiming over 700 000 lives as of 2019. With an estimated annual rate of 9 deaths per 100 000 population, suicide is responsible for 1.3% of all deaths.[1] Although it has been established that certain groups of the population are at a higher risk of suicide, such as young men aged 15–29 years old and individuals with affective disorders or substance abuse, no person is exclusively protected from suicide.[2]

Elite athletes are often regarded as the healthiest individuals in the general population due to their physical fitness and particular attention to many aspects of their health. Nonetheless, these individuals could also be at risk of suicide. Although physical exercise is associated with beneficial effects on mental health,[3] athletes are exposed to unique risk factors that might contribute to maladaptive coping behavior and depressive disorders potentially leading to a higher risk of suicidality.[4] Increasing pressure is often experienced by athletes since their career usually lasts only a few years on average. Therefore, mental health issues may be downplayed for the sake of short-term success.[5–8]

The news reports regarding suicide deaths of prominent sports figures raised awareness among the medical and general population regarding this issue. Professional athletes may live in a distinct environment and could carry unique features (e.g., frequent anxieties due to performance, overtraining due to unhealthy competition) that place them in a more vulnerable mental health position, which can further trigger a suicide attempt.

Many scientific works have attempted to explore risks and protective factors for suicide among athletes, but many questions remain unanswered. While many studies reported lower rates of suicide among elite athletes, others contradicted by presenting opposite findings. The association between previous injuries, particularly head trauma leading to traumatic brain injury (TBI) and chronic traumatic encephalopathy (CTE), has been investigated thoroughly with somewhat conflicting results.[9] Similarly, the relationship between performance-enhancing drugs and anabolic medications has been a matter of dispute among researchers.[10] Lastly, the rates of affective disorders increasing the risk of suicide among athletes may be underestimated or underreported due to stigma and lack of awareness in this population.[4]

There has been a great interest in the topic of suicides among athletes, which led to a large number of published articles. This work therefore aims to explore epidemiology, risk factors, protective characteristics as well as potential screening and preventive measures specific for suicide among athletes based on a comprehensive review of the recent literature.

2 | METHODS

A comprehensive literature search was carried out through PubMed and Embase databases for relevant publications (Figure 1, Table 1). No formal restrictions were applied regarding the timeline and age of participants. However, studies were limited to original research studies as well as systematic reviews and meta-analyses published in English and German languages from the day of inception through March 2022. Selected practice guidelines and consensus statements from international professional organizations in the fields of psychiatry, psychotherapy, and sports medicine were also included in the discussion of this work. Narrative reviews, editorials, commentaries, communications, and op-eds were considered only for the reference of studies to be potentially included. Citation lists of each study were additionally screened to identify any possibly missed items. Emphasis was given to the works published in the previous 15 years. This detailed approach provided large search coverage. Various combinations of the following search terms were applied: "athletes," "elite athletes," "elite sports," "professional sports," "sportsmen," "sportswomen," "high-performance athletes," "high-performance sports" matched with "suicide," "suicidal ideation," "suicidality," "suicide attempt," "suicide rate," and "suicide risk." Articles including high-school sports participants only were excluded. Studies were selected based on conceptual relevance, practical applicability in the field of psychiatry, research novelty, and methodological strength. The search process was carried out by two authors (E-M.P. and S.E.), and decisions were made based on consensus.

3 | SUICIDALITY AND SUICIDES IN ELITE SPORTS

3.1 | Incidence and prevalence

Several studies assessed the incidence of suicide among sports professionals in comparison with the general

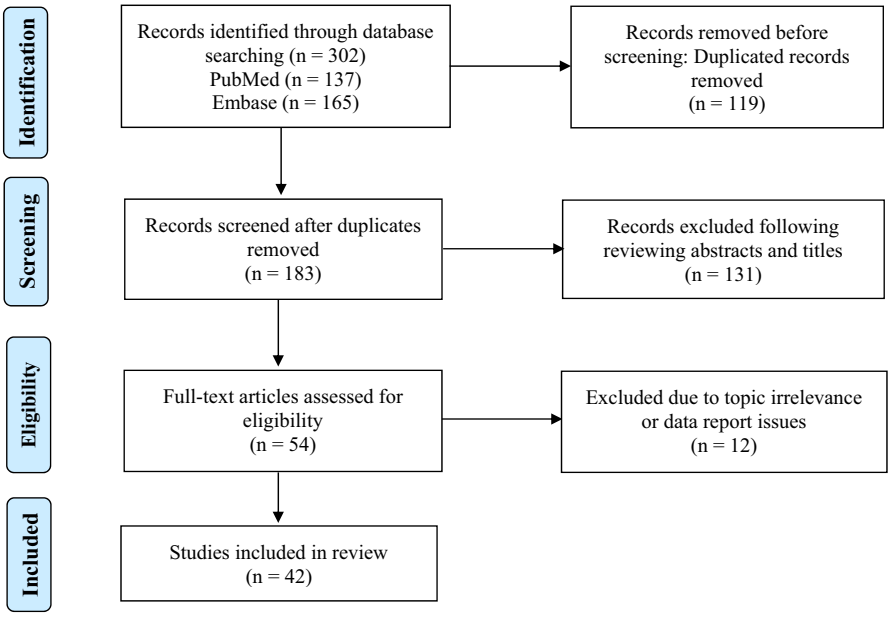

FIGURE 1 Flow diagram of study selection (PRISMA flow diagram).

population. In general, the rates usually appear to be lower or at least similar in athletes from various professional levels and from different sports disciplines regardless of their current professional status (i.e., active vs. retired).[11–23]

The suicide rate among college athletes from National Collegiate Athletic Association (NCAA) was estimated to be 0.8–0.93 per 100 000 a year representing the second to the fourth leading cause of death (following traumatic accidents, cardiovascular deaths, and homicide) with a total of 7.3%–17% of all-cause mortality in this population group.[14,23] Suicide rates among collegiate athletes were thus shown to be lower among players in comparison with the general population and non-athlete college students.[14,23] Equal rates of suicide attempts (1.3% vs. 1.6%, $p < 0.01$) and lower rates of suicidal ideation (6.4% vs. 11.3%, $p < 0.01$) between athlete and non-athlete students were reported among 165 210 respondents to the National College Health Assessment survey.[21] In the longitudinal analysis of this survey from 2011 to 2019 involving over a half of million responders from 607 institutions, collegiate athletes showed consistently lower rates of mental health-related symptoms and diagnoses (including suicide) compared to non-athletes (with an exception of addiction and eating disorders), although rates increased in both groups during the study period (suicide rates rose from ~2% to ~3%).[22] Moreover, academic performance was less likely to be influenced by mental health problems in athletes vs non-athletes.[22] The reported increase in suicidal intention and attempts may be attributed to emotional exhaustion seen as 85% of college students.[24]

Similar trends were seen in suicide rates among professional (non-collegiate) athletes. Lemez and colleagues reported lower suicide rates among North American active professional athletes in comparison with the general population.[17] Suicide was the fourth leading cause of death after accidents, homicides, and cardiac diseases, accounting for 6.8% of all deaths.[17] A study by Duncombe et al.[18] found that former US Olympians had a 32% lower risk of death from mental disorders, eating disorders, and suicide (standardized mortality rate [SMR] = 0.68, [95% CI 0.49–0.91]). Likewise, with an average follow-up of 50 years Kettunen et al.[19] found similar rates of suicide between Finnish non-athletes and elite athletes, the latter demonstrating on average 5–6 years longer life expectancy. Lastly, a meta-analysis of over 27 thousand male athletes found that lower rates of suicide among former national football league (NFL) players (standardized mortality rate: 0.54, 95% CI: 0.37–0.78, $p < 0.001$) and former or active soccer players (standardized mortality rate: 0.55, 95% CI: 0.46–0.67, $p < 0.001$).[15]

Former elite athletes also do not appear to be at a higher risk of death by suicide. A study involving a cohort of over 3400 former National Football League players who had played 5 and more years showed an even lower rate of suicide compared to the general US population (SMR = 0.47; 95% CI: 0.24–0.82).[25] No difference in the risk of suicide was found in a large cohort of 7676 former soccer players, which, however, had a decreased likelihood of hospital admission due to stress-related disorders, depression, anxiety, substance abuse, and bipolar disorders.[26]

Certain distinct features can be seen across all levels of athletic profession. For example, male collegiate athletes were 3–4 times more likely to commit suicide compared to their female counterparts, accounting for up to 83% of all deaths by suicide in collegiate atheletes.[14,23] One study by Maron et al.,[12] however, reported a higher frequency of suicide among young female competitive athletes (13 of 253 [5.1%] deaths vs. 53 [2.5%] of 2153 deaths) using a

TABLE 1 Descriptive characteristics of included studies

Study ID (country)	Sample size	Participants	Study type	Outcome
Du Preez 2017 (Australia)	278	Current elite rugby league players	Cross-sectional	Lower prevalence of depression High prevalence of GAD and alcohol misuse
Foskett 2017 (UK)	143	Elite athletes	Cross-sectional	47.8% had signs of anxiety/depression 26.8% had signs of distress Career dissatisfaction as a significant independent predictor of depression and anxiety
Gouttebarge 2016 (The Netherlands)	204	Elite Gaelic athletes	Prospective cohort	23% of prevalence for adverse alcohol use; 48% for anxiety/depression 11% of 6-month incidence of sleep disturbance and 21% for anxiety/depression
Gruber 2000 (USA)	75	Female bodybuilders or fitness athletes	Cross-sectional	33% had current or past AAS use 56% of the users had hypomanic symptoms 40% had depressive symptoms during withdrawal
Johnson 1997 (Sweden)	81	Long-term injured elite athletes	Cross-sectional	Women were more anxious and tense and used emotion-focused coping strategies; team-sport athletes accepted easily help from other
Nixdorf 2016 (Germany)	199	German junior elite athletes	Cross-sectional	Athletes in individual sports had higher depression scores
Silver 2001 (USA)	5034	Residents of New Haven, CT	Cross-sectional	Head injury was associated with for major depression, dysthymia, panic disorder, OCD, phobic disorder and drug abuse/dependence, suicide attempt, and suicide
Waddington 2006 (England)	2863	Professional soccer players (6% retired)	Cross-sectional	6% knew players who used performance-enhancing drugs, and 45% of players knew players who used recreational drugs
Prinz 2016 (Germany)	157	Former female soccer players	Cross-sectional	32.3% of all-career prevalence of depression symptoms
Pope 1994 (USA)	160	Currently active athletes	Case–control	23% of steroid users had major mood disturbances
Ostovar 2017 (Sweden)	453	Current bodybuilders	Cross-sectional	Prevalence of major depressive disorder 19.7%; bipolar disorder 3.8%; generalized anxiety disorder 16.6%; 33% of performance-enhancing drug abusers had acute psychological disturbances; 26.7% overall prevalence of psychiatric disorders
Mez 2017 (US)	202	Deceased former football players	Case series	In mild CTE, the most common cause of death was suicide (27%), while neurodegenerative disorders in in severe CTE pathology (47%)
McKee 2013 (USA)	85	Subjects with history of repetitive mild TBI	Case–control	26% had suicidal ideation and 14% completed suicide
Madsen 2018 (Denmark)	34 529	Individuals who died by suicide	Retrospective cohort	TBI was associated with higher risk of suicide
Junge 2018 (Germany)	290	Current female elite soccer players	Cross-sectional	Rates of anxiety and depression disorders were similar to general population
Junge 2016 (Switzerland)	10	Current professional soccer players	Cross-sectional	Similar prevalence of depression as the general population; while younger male players had a higher prevalence of depression
Gouttebarge 2017 (The Netherlands)	203	Current elite athletes	Prospective cohort	6% had adverse alcohol use; 57% had symptoms of anxiety/depression
Dudek 2015 (Poland)	490	Current extreme or high-risk athletes	Case–control	Higher rates of bipolarity and impulsivity

TABLE 1 (Continued)

Study ID (country)	Sample size	Participants	Study type	Outcome
Cottier 2011 (USA)	644	Retired NFL players	cross-sectional	52% used opioids during and 71% misused; three times of the rate in general population; 15% of misusers continues to do so; significant pain, undiagnosed concussions, and heavy drinking were predictors of abuse
Beable 2017 (New Zealand)	370	Currently active athletes	Cross-sectional	21% had depressive symptoms; young age, retirement phase, individual sports were risk factors
Gorczynski 2017	3386	High-performance athletes	Systematic review	Similar rates of depression as in the general population
Rice 2016	18 323	Elite-level athletes	Systematic review	Similar rates of anxiety and depression as in general population
Moreland 2016 (USA)	8603	Collegiate athletes	Systematic review	Availability, peer norms, stigma, gender influence utilization of mental health services
Sun 2020 (China)	774	Currently active athletes	Cross-sectional	Psychological strain was associated with suicidal ideation, hopelessness, and depression
Piacentino 2015	4007	Current and former athletes	Systematic review	AAS abuse and dependence were associated with psychiatric disorders
Hussain 2022 (Pakistan)	60	Currently active athletes	Cross-sectional	AAS abuse was linked to suicidal ideation, interpersonal problems, depression, hostility, paranoia, psychoticism
Parssinen 2000 (Finland)	62	Male competitive powerlifters	Retrospective case control	Suicide was the common cause of death
Fralick 2018	7 million	General population	Systematic review	Concussion was associated with suicide attempt, suicidal ideation, and twofold higher risk of suicide
Manley 2017	2269	Current and former athletes	Systematic review	Multiple concussions were linked to cognitive impairment and mental health problems
Wortzel 2017	17	Deceased athletes	Systematic review	No links between CTE and suicide
Maroon 2015	153	Deceased athletes	Systematic review	No links between CTE and suicide
Constanza 2021 (Italy)	2	Retired athletes	Case reports	Both died from suicide
Ojio 2021 (Japan)	251	Current male rugby players	Cross-sectional	32.3% had symptoms of mild anxiety and depression; 4.8% and 5.2% had moderate and severe symptoms; 7.6% had suicidal ideation
Armstrong 2015	20 189	Current or former athletes	Systematic review	Student athletes were less likely to be depressed compared to their non-athletes
Khader 2020 (USA)	113 185	Student athletes	Cross-sectional	Sleep disturbances were linked to suicidal ideation
Morales 2021	27 477	Former and current professional male athletes	Systematic review	Lower risk of suicide and mental health problems compared to the general population
Timpka 2020 (Sweden)	402	Elite athletes (28% retired)	Cross-sectional	1 in 6 international athletes experienced suicidal ideation linked to sexual abuse and lower coherence in women and behavioral disengagement in men
Schaal 2011 (France)	2067	High level or junior athletes	Cross-sectional	Female athletes had higher rates of anxiety, eating disorders, depression, sleep problems, and self-harming behaviors

(Continues)

TABLE 1 (Continued)

Study ID (country)	Sample size	Participants	Study type	Outcome
Kokotailo 1996 (USA)	1046	Student athletes	Cross-sectional	Women were more likely to attempt suicide and have riskier sexual behavior. Aggressive driving, alcohol use was more common in men
Jovanovic 2022 (Slovenia)	97	Elite athletes	Cross-sectional	Male athletes were more impulsive and more likely to abuse alcohol and other psychoactive substances impulsive and risk-taking. Female athletes were more likely to develop eating disorders
Tran 2020 (USA)	39 840	Student athletes	Longitudinal	Higher risk of depression and suicide among Asian and Pacific Islander students
Anchuri 2019 (USA)	165 210	College student athletes	Cross-sectional	Difficulties with social relationships were linked with suicide attempt
Dodd 2021 (USA)	153	College student athletes	Cross-sectional	Greater capability for suicide compared to the general population
Edwards 2020 (USA)	502 780	College student athletes	Longitudinal	Similar rates of suicide and suicidal ideation compared to non-athletes. Increased rates of mental problems during study period.
Maron 2014 (USA)	4 million	College student athletes	Longitudinal	Lower rates of suicide compared to the general population
Rao 2015 (USA)	4 million	College student athletes	Longitudinal	Lower rates of suicide compared to the general population
Kettunen 2014 (Finland)	4020	Former elite athletes	Longitudinal	Similar rates of suicide compared to the general population; higher rates of suicides among power sports athletes
Shah 2016 (UK)	2794	Current and former elite cricketers	Retrospective	Similar rates of suicide compared to the general population
Lehman 2016 (USA)	3439	Former NFL players	Longitudinal	Lower rates of suicide compared to the general population
Lindqvist 2013 (Sweden)	1199	Former elite athletes	Longitudinal	Increased mortality rates and suicide related to AAS use
Russel 2020 (UK)	30 704	Former professional soccer players	Longitudinal	Lower rates of mental health problems requiring hospitalization and similar rates of suicide compared to the general population.
Duncombe 2020 (USA)	8124	Former US Olympian athletes	Longitudinal	Lower risk of death due to mental disorders and self-harm
Konietzny 2018 (Germany)	159	Professional athletes with back pain	Cross-sectional	Pain levels were associated with a higher risk of depression and suicidal ideation
Lemez 2016 (Canada)	205	Deceased athletes	Retrospective	Suicide was the fourth cause of death among athletes
Maron 2016 (USA)	2406	Deceased athletes	Retrospective	Higher frequency of suicide among females compared to males
Nguyen 2019 (USA)	6127	Current and former NFL and NBA players	Longitudinal	Similar rates in suicide between NFL and NBA players
Webner 2016 (USA)	26 702	Current and former athletes	Longitudinal	Common life stressors (such as retirement from sport, loss of steady income, divorce, failed business ventures, estrangement from family members and medical, psychiatric and/or substance abuse problems) were associated with suicide among athletes
Lester 2014	4814	College students	Cross-sectional	A reduced incidence of suicidal ideation with college sports participation

Abbreviations: AAS, anabolic-androgenic steroids; CTE, chronic traumatic encephalopathy; GAD, generalized anxiety disorder; NBA, national basketball association; NFL, national football league; OCD, obsessive compulsive disorder; TBI, traumatic brain injury.

30 years analysis of the US National Registry of Sudden Death in Athletes. No significant racial or ethnic differences have been noted in the mentioned studies.[14,21,23]

Some types of athletic activities have been associated with higher tendencies of suicide. Power-sport[19,27,28] and firearm[18] shooting participants were shown to have increased risk and incidence of suicides compared to both other athletes and the general population. These findings are supported by an analysis by Dodd et al.[29] showing an association between contact sports with a fearlessness about death. Compared to collegiate athletes participating in soccer, track/cross-country, baseball, and swimming, a higher incidence of suicide was identified in American football athletes (RR 3.27; [95% CI: 1.65–6.5], $p < 0.01$).[14]

Some studies have found geographic differences (e.g., urban vs rural), which can be largely attributed to cultural and societal aspects of life and health practices in various countries.[16,18–20,26,27] Differences also exist regarding the age of athletes, which again corresponds to results determined in the general population. For example, in the United States, suicide was the second cause of sudden death from non-cardiovascular causes among collegiate athletes accounting for 17% of mortality based on an analysis of data from the 10-year study period from 2002 to 2011.[23] Suicide was found to be the fourth major mortality cause among North American athletes[12] and third among former power sports athletes from Sweden.[27]

These findings collectively support the general notion of athletes being at least as mentally healthy as others in the population, if not healthier. It has been suggested that a higher level of physical activity and team sports participation may lead to beneficial outcomes, such as better mental health, self-esteem, and overall satisfaction with life.[19,30,31]

3.2 | Risk factors

Risk factors for suicide are formally divided into environmental and individual levels,[32] where psychiatric disorders play a primary role in the latter.[33] Athletes can be affected by a wide range of mental health problems, broadly similar to those experienced by the general population.[10,34] However, risks specific to sports and the occurrence of mental illness in sports should also be taken into account. There are discrete precipitating and predisposing factors common in athletic environments, whereas others are rare.

For example, the mental burden among professional and elite athletes might be often enhanced through pressure to succeed, unsatisfactory performance, competitive failure, and career dissatisfaction.[35] The severe injury necessitating surgery has been discussed as especially life-threatening when it comes to suicidal ideations and behavior because it demands an extended rehabilitation process that might lead toward reduced athletic skills despite adherence to rehabilitation, a perceived lack of competence upon returning to sports when compared with preinjury levels and being replaced by a teammate at their given position.[36] Chronic and recurrent musculoskeletal problems, and overreaching and overtraining may be detrimental for athletes' mental health.[34]

It has been estimated that almost half of sports professionals (46.4%) had at least one symptom of a mental disorder,[37] and there is a roughly 50% of lifetime prevalence of mental health problems (58.2% for females and 42.3% for males).[38] Certain determinants such as negative life events or injury increase the vulnerability of athletes to mental health problems.[37,39] According to the Interpersonal-Psychological Theory of Suicide, individuals that died by suicide experienced repeated exposure to painful and provocative situations.[40] The habituation of physiological pain and a diminished fear of death promotes a higher likelihood of suicide.[41] Further common contributors may include toxic coaching and team environment, poor performance, or fatigue.[42,43] Low social support is often cited as another general risk factor for developing mental illness.[35] Issues related to suicidality, such as risky behavior and impulsivity, are often practiced in high-performance sports.[40]

Retirement is a complex and difficult period in the lives of professional athletes. Most suicides happen after active phase of the career (~16 years after retirement) and in the context of substance abuse and/or a serious life event in the previous year, such as health concerns, financial difficulties, or relationship disharmony.[16,17,20] Disempowerment is often perceived by athletes at the end of their career, especially when it happens abruptly due to injuries.[44] The consequences of retirement on self-confidence and body image might lead to a mental crisis.[45] Notably, certain physical factors, such as declining performance or injuries, can also negatively affect self-perception and self-esteem of athletes leading to narcissistic crises.[36,46]

4 | SPECIFIC MENTAL HEALTH PROBLEMS AND ILLNESSES

4.1 | Depression and anxiety

Depression is present in more than half of the suicide cases.[47] Most studies report a similar prevalence and gender distribution of depression among athletes and non-athletes. A comparative meta-analysis of five international studies, including data from 1545 high-performance athletes and 1811 non-athletes, revealed rates of mild or more

severe depressive symptoms ranging from 3.7% to 26.7% for male high-performance athletes (vs. 7.6%–34.4% for non-athlete males) and from 9.8% to 36.5% for females (6.1%–42.5% for non-athlete females), did not to find any statistically significant differences compared to non-athletes.[48] Similar rates were found in other investigations, not included in that analysis, indicating depression as being the most common mental illness in this group of the population, with a predominance in females,[35,37,49–58] which were 50% more likely to have mild to severe depressive symptoms.[48] A systematic review of 11 studies by Armstrong and colleagues evaluating depression frequency among student-athletes concluded that collegiate athletes had a lower likelihood of experiencing depression compared to non-athletes due to some protective factors, which can include elevated self-esteem and strong social networks. Importantly, the authors also pointed out that student-athletes tend to seek help more rarely.[59] Career dissatisfaction was seen as an independent risk factor for anxiety and depression in a cross-sectional survey of elite athletes from the UK.[49] The differences between reported incidences can be accounted for by heterogeneity in methodology, including sampling, assessment tools, and applied criteria for depression.

Although prevalence seems to be proportional among various disciplines of athletic sports, individual sports athletes are twice as likely to suffer from depression or anxiety as athletes in team sports, possibly indicating the protective nature of a team supportive environment.[39,60] Team-sports athletes were shown to accept help and support more readily.[61] Performance failure was a more significant factor for developing depression among individual sports.[62] Furthermore, elite athletes in comparison with non-elite counterparts are more prone to depressive and anxiety episodes, where performance failure is perceived more dramatically.[63,64] Positive correlations albeit not statistically significant were shown between level of division in National Collegiate Athletic Association (NCAA) and suicide death rates.[14] Contrary to these findings, other studies reported lower depression scores with higher levels of play.[39,55,65]

Average depression scores may also differ depending on the playing position. In soccer, for example, forwards, goalkeepers and athletes playing in multiple positions were shown to have comparably higher rates of depression compared to other positions.[55,65] Prinz and colleagues reported that female elite soccer players stated that "conflicts with coach/management" and "too little support/acknowledgement by the coach" were fundamental reasons for their lows in mood, while only a small percentage described the coach as helpful in coping with lows in mood.[55]

The time point in a sports season, phase of the competition, training load, and stage in career were identified to influence the prevalence of depressive symptoms as well. Players tend to be more depressed during pre-season period vs in-season (12.6% vs. 10.1%).[63,66] Furthermore, former athletes have a higher tendency of symptoms of depression compared to active professionals (39% vs. 26%).[35]

Although athletes have a similar incidence and prevalence of depression when compared to the general population, etiology and contributing factors might often differ. Chronic pain, known to increase the risk of depression with a 5%–15% lifetime prevalence of suicide attempts,[67] is seen in as many as half of retired American football players.[52] Severity of depression can be exaggerated by elevated levels of pain, stress, and the so-called "pain-related thought suppression" (characterized as inhibition of thinking about pain).[68] Sports injuries are shown to have a timely association with depression and anxiety incidences in athletes.[37,69] It is important to note that a psychological strain, frequently experienced by sports professionals, facilitates suicidal ideation. This relationship is further mediated by feelings of hopelessness and depression.[70]

Suicide attempts are twice as likely in people with anxiety disorders, while concurrence with depression may increase the suicide risk as high as by 17-fold.[71] In this context, it should be noted that anxiety is a common finding in athletes,[35] often perceived as a facilitator for the performance related to anger, specific self-esteem strategies, and patterns of perfectionism.[72–75] Generalized anxiety disorder has been reported in approximately 6%–8% of athletes with female predominance,[37,51,60,76] which is also corresponding to the general numbers observed among non-athletes.[77,78]

4.2 | Mania and bipolar disorder

The recognition and diagnosis of bipolar disorder, which is seen in 1%–2% of the general population,[79] third to half of which attempt suicide,[80] represent major challenges in athletes, who often share periodic patterns of impulsivity, high self-confidence, elevated levels of activity, as well as depressed mood or lack of motivation. Moreover, a peak in the athletic career performance and the usual age of bipolar disorder onset co-occur during early adulthood.[81] Thus, the exact prevalence of the bipolar disorder among athletes is not known.[82] Dudek et al.[83] reported in a web-based study that athletes engaging in extreme and high-risk sports had significantly higher scores of bipolarity in the Mood Disorder Questionnaire than controls. The impact of sport, in combination with performance-enhancing drugs on the occurrence of bipolar, was investigated in two studies. Among 41 anabolic-androgenic steroids (AAS) using bodybuilders and football players, 23% displayed symptoms of bipolar disorder.[84] Ostovar

et al.[85] reported the prevalence of bipolar symptoms in bodybuilders using performance-enhancing drugs at 3.8%.

4.3 | Substance abuse

Individuals with substance abuse are at least twice as likely to attempt suicide as others.[86] Elite athletes are often at risk of being exposed or predisposed to addictive substances leading to increased morbidity and mortality. In fact, a significant increase in drug use by professional athletes since the 1960s derives from the fact that 45% know players who are using recreational drugs.[87,88] Regular scrutinous drug testing among athletes may be preventive in using illicit drugs; nonetheless, eight percent reported using an illicit drug in the past year.[89] Opioids use, often initially prescribed for chronic pain management, appeared to be three times more common among NFL players (with a current prevalence of 7%), where half of the athletes (52%) reported to have used opioids during their career, of which 71% misused at some point and 15% abusing currently.[90]

Alcohol use among athletes is thought to have a binge nature of consumption, which usually happens during holiday breaks.[91] Especially in team sports like rugby, alcohol consumption seems to be part of the cultural context of the sport.[92] Moreover, teams and individual players are often sponsored by alcoholic beverage companies resulting in higher alcohol consumption.[93] Many athletes might use alcohol to overcome anxiety and pressure.[94] The prevalence of adverse alcohol use was reported to be 23%.[58] Alcohol abuse leads to disinhibition, impulsiveness, and impaired judgment, which may boost suicidal behavior.[95] Professional athletes may often continue illicit drug consumption after their career, which may be unreported.

4.4 | Performance-enhancing drugs

According to one previous report, 6% of professional soccer players in England know other players are taking performance-enhancing drugs,[87] among which AAS represent the most prominent group. AAS usage during professional career years was linked to suicidal ideation and suicide in several reports depicted by violent, impulsive, aggressive, and disinhibited behavior.[28,96–100] Among AAS abusers, the proportion of suicide from all deaths varied from 21%[99] to 37.5%,[28] which was almost three times higher compared to the general population.[28,98] Importantly, female athletes can also abuse AAS. A small study of 75 bodybuilders and weightlifting athletes found that a third of participants used it in the past or were using it at the time of evaluation.[101] Concomitant use of other substances, which is not an infrequent scenario among AAS users, may also enhance the risk of suicide in athletes,[99] whereas low income and unemployment (common risk factors for suicide)[102] may become a risk for AAS misuse among athletes.[100]

4.5 | Traumatic brain injury

Head trauma and specifically TBI represents a major issue in athletes, since more than 1.3 million new cases are registered in the European population every year[103] and almost a third of all TBIs are estimated to be sports-related, where 3 out of 4 events occur in males and adolescents and young adults share highest incidence.[104]

A systematic review and meta-analysis of 17 studies encompassing data of over 7 million individuals in total found that concussion or mild TBI was linked to a doubled risk of suicide (relative risk, 2.03 [95% CI: 1.47–2.80]; $p < 0.001$), whereas concussion was additionally associated with higher likelihood of suicide attempt and suicide ideation.[105] It should be noted, however, that the actual risk for completed suicide was still less than 1% (0.5%–0.59%), while many of the included studies were not able to control important factors (e.g., childhood experiences, family history of suicide, substance abuse, and major life stressors).

A retrospective cohort employing a national patient registry in Denmark including data of over 7 million individuals, of which half of million people with TBI, indicated a nearly twofold greater incidence rate ratio of suicide with any degree of TBI (1.9, [95% CI: 1.8–2.0]).[106] A higher incidence was associated with greater severity of TBI (concussion: 1.8, [95% CI: 1.7–1.9]; structural brain injury: 2.4, [95% CI: 2.2–2.6]).[106] Although the suicide risk was highest during the first half-year following TBI, it remained increased for a minimum of the next 7 years.[106] Moreover, having more than one TBI event further increased the risk of suicide.[106]

Another study by Silver et al.[107] based on surveys of 5034 people from general population found that any history of head injury resulting in a loss of consciousness or confusion was linked to an increased lifetime incidence of psychiatric diagnoses. After adjustment for sociodemographic factors, use of alcohol, and quality of life, participants with head injuries had higher risk for major depressive disorder, dysthymia, panic disorder, phobic disorder, obsessive–compulsive disorder, and illicit drug abuse or dependence.[107]

Recurrent concussions were also shown to have an almost linear relationship with the risk of later life depression episodes (i.e., higher risk of depression with a higher number of concussion), independent from physical

performance and years since sports retirement.[108] Once they happen, concussions are not only more likely to re-occur, but they tend to be more frequent and severe.[109] These findings are supported by a reported increased susceptibility of retired professional contact sports athletes to diminished cognitive function over time, where multiple concussions can increase the risk of cognitive impairment and mental health problems and illnesses.[110]

The path from brain injury to psychiatric alterations is not completely understood. It has been described that TBI-induced lesions might lead to biochemical alterations, such as an increase in excitatory neurotransmitters and a subsequent neuronal loss, which has been linked to chronic depression.[109] Due to frontal lobe injury, TBIs are hypothesized to play a role in impulsivity and disinhibition.[111] Nevertheless, neuroimaging modalities involving magnetic resonance imaging (MRI), positron emission tomography (PET), and diffusion tensor imaging (DTI) indicated various structural alterations among former professional football players. However, these brain changes were neither consistent nor specific, limiting imaging use and interpretation in this context.[110]

4.6 | Chronic traumatic encephalopathy

Chronic traumatic encephalopathy is currently defined as a neuropathological entity characterized by cumulative neuropsychologic deficits in association with recurrent concussions or TBI.[112] The diagnosis is established post-mortem with identification of abnormal accumulated hyperphosphorylated tau protein (p-tau) within neurons, astroglia, and perivascular spaces of cortical sulci.[112] The links between CTE and suicide remain unclear, and a matter of ongoing research with multiple studies attempted to clarify these links without significant major yet breakthroughs.[4,113]

A systematic review of 153 pathology case reports found links between mild TBI and CTE in contact sports athletes, but no significant associations between CTE and suicide. Moreover, no correlations were identified between CTE and age of death.[114] In some individual case studies and series, suicide was observed in persons with CTE, including but not limited to athletes.[115–120] An analysis of 85 brain specimens from former athletes, military veterans, or civilians with histories of repetitive mild TBI found CTE in 80% of the cases with 26% experiencing suicidal ideation and 14% of deaths occurring due to suicide and 12% due to substance abuse.[121] Mez et al.[122] reported that 87% of 202 deceased American Football Players were diagnosed with CTE, 96% of them featured psychiatric conditions during their lifetime, including depression (67%), anxiety (52%), impulsivity (89%), substance abuse (67%), and suicidality (56%). Authors found that the most common cause of death for participants with mild CTE pathology was suicide (27%), whereas for severe CTE pathology it was neurodegenerative disorders (47%).[122] Contrary to these findings, Bieniek and colleagues could not find differences in clinical conditions between CTE positive and negative subjects, although CTE was present in 31.8% of former athletes with a history of repetitive brain trauma.[123]

The overall evidence for CTE in relation to suicide remains largely scarce and limited to low quality reports. The associations are further questioned by lower rates of suicide among athletes.[16,25] Many questions exist with regard to the homogeneity of CTE definition and diagnosis. Future studies employing longitudinal study design and decent sample size along with pre-defined commensurate criteria are necessary to further explore and expand our current understanding of the relationship between repetitive head trauma in sports, the development of CTE as a result, and possible clinical manifestations, including psychiatric conditions that may lead to suicide.

4.7 | Other common individual risk factors

Smith and Milliner identified several further important factors enhancing suicidality in athletes, such as an injury necessitating surgical intervention, a long rehabilitation period restricting sports participation, deterioration in the athletic skills and competence, and being replaced by other players, where younger age further increased the risk. Sustaining an injury following considerable athletic success can be another risk factor for suicide.[36] Physical injury was found to be correlated with depressive mood as well.[61] Dodd et al.[29] found that fearlessness about death was positively correlated with the number of sports-related injuries and subjective pain tolerance, the latter being higher in athletes than in non-athletes. Considering that suicide attempts are often carried out after precipitating events, a greater risk for suicidal behavior caused by sport-specific stressors may exist among professional and elite athletes.[124]

In a national survey, including over 31 thousand collegiate athletes, Khader et al.[125] showed an association between sleep problems (seen in 24%–36% of college students) and suicidal ideation. Interestingly, more nights of insufficient sleep per week were linked to higher risk. Another prospective cohort study by Gouttebarge et al.[58] found 11% of the 6-month incidence of sleep disturbance among 204 elite Gaelic athletes.

A study by Timpka et al.[126] conducted on 192 Swedish athletes found that the prevalence of suicidal ideation

was 15.6% (men 17.4%; women 14.2%). Suicidal ideation among women was linked with sexual abuse and a lower sense of coherence.[126] Noticeably, being sexually abused in the past was also linked with non-sports injury prevalence and participation in an endurance event.[126] Among men, suicidal thoughts were associated with the use of behavioral disengagement for coping.[126]

In a cohort of over 26 000 professional American football players from 1990 to 2015, Webner and Iverson reported that 80% of individuals who committed suicide experienced life stressors, such as retirement, loss of steady income, divorce, failed business ventures, problems with family members, as well as medical or psychiatric disorders.[16] Difficulties with social relationships were strongly associated with suicide attempts among collegiate athletes.[21] These findings indicate the need for coping and protective strategies for athletes when they find themselves in critical circumstances.

Several studies investigated differences in suicide and associated risk factors among female and male athletes. A study by Schaal et al.[127] reported a higher prevalence of psychopathologies in female high-level athletes compared to male counterparts (20.3% vs. 15.1%) vastly attributed to eating disorders, depression, sleep problems, and self-harming behaviors. Johnson et al.[61] also indicated a comparably higher level of anxiety among female athletes. Interestingly, female athletes also used emotion-focused coping methods more frequently.[61] Risk-taking behaviors, commonly seen among college students, also may differ between males and females.[61] Male collegiate athletes were also shown to have higher rates of alcohol use and binge drinking as well as the use of tobacco, marijuana, or cocaine.[128,129] Female athletes tend to develop eating disorders at a higher rate,[128] while male athletes have more impulsive and risk-taking behavior.[128,129]

Lastly, a longitudinal study by Tran described trends in racial and ethnic differences among 39 840 American student-athletes in relation to suicide and associated risk factors.[130] The author noted increased risks of depression and suicide among college athletes of Asian or Pacific Islander or multiracial background in comparison with White athletes.[130] The significance of these findings should be further evaluated in other populations considering the cultural and demographic features of a studied region.[130]

5 | CLINICAL IMPLICATIONS AND RECOMMENDATIONS

Findings from the reviewed literature indicate that despite loud media coverage and increased public attention, elite athletes do not appear to be at a higher risk of suicide compared to other groups of the population. In fact, most athletes show lower rates of mortality related to suicidality. Nevertheless, no one is immune to this crucially important health problem, and certain risk factors, such as TBI, physical injury, or AAS use, should be considered when assessing mental health issues among athletic professionals. Although the most common risk factors tend to be similar in this population group (i.e., affective disorders and substance abuse), the predisposition for these issues eventually leading to suicide may differ considerably (Table 2).

A summary of most relevant risk factors for suicide in elite athletes was published by International Olympic Committee in 2019, which encompassed a history of childhood trauma, anxiety, agitation, aggression, impulsivity, hopelessness, interpersonal conflict, drug and alcohol use, physical illness/injury, prior suicide attempts, and sleep disturbance.[131] Identification and mitigation of the reviewed risk factors are highly recommended.[131]

Despite similar rates in depressive symptoms and disorders, special attention should be paid to those athletes with substantial injuries, decreasing sports performance, or chronic pain. The latter may frequently co-appear with misuse of opioid painkillers.[90] The context in which suicidality is assessed is important, where some features, including team sports vs individual sports, playing position, type of athletic sports, the time point of the season, phase of the competition, mental health intersect, performance and career satisfaction, training load, recent or concurrent injuries, physical or sexual abuse, and stage in career, might mediate the suicide risk.[61,131] Since most suicides occur in non-active athletes, long-term follow-up of mental health concentrating on issues related to physical health problems, social isolation, financial instability, or tensions, as well as the process of transition from sport in private life is required.[131]

TABLE 2 Suicide risk factors for elite athletes

Depressive disorders[35,37,48,51–57]
Anabolic-androgenic steroids abuse[28,96–100]
Illicit drug abuse[87,88,90,91,95]
Traumatic brain injury[106]
Retirement from athletic activity[16,17,20,35]
Physical injury necessitating surgical intervention[36,37,69]
Chronic pain[52,67,90]
Life stressors (related to family, finance, society, business)[16,17,20,21]
Physical or mental illness[36]
Anxiety disorders[35,37,51,60,72–76]
Sport-specific environmental stressors[124]
Lack of support, feeling hopelessness[55]

Current evidence shows an increased risk of suicide with a past TBI episode.[106] Further research should therefore investigate the potential usefulness of assessment of previous TBI as a part of history taking while evaluating patients for suicidality.

Performance-enhancing drugs and AAS may be frequently abused, elevating the risk of suicide.[28,98] Although periodic testing for medication use is an effective barrier to this issue, a medical evaluation should seek to identify potential misuse of AAS, as concurrent use of other substances, including illicit drugs, is common.[99] Clinicians should bear in mind that alcohol abuse carries a binge character[91]; thus, an absence of current signs of alcoholic intoxication should not discourage further interception of the disorder. Higher vigilance is prompted for recognition characteristic signs and symptoms of substance abuse. Educating players by providing balanced and objective information about potential risks and benefits were shown to improve understanding of consequences and adverse effects of substance abuse.[132]

Specific cultural and country-related issues should be recognized and addressed accordingly. For example, firearms were reported to be the most common instrument of suicide among American athletes[14,16] and Australian elite cricket players.[20] Opioid prescription and availability practices in various parts of the world may also play a significant role in this context,[90] and thus should be taken into account. Both mental healthcare providers and public health experts, as well as policy makers, should adopt strategies in minimizing the risk of associated harm.

Encouragement of help-seeking for mental health issues may decrease suicide risk. More than half of athletes (57.1%) reported to seek for professional help for their psychological problem, but only 4.5% visited a psychiatrist.[37] In a German study with 290 first and lower league female soccer players, only a third of the players who stated that they currently wanted or needed psychotherapeutic support received it.[57] In a study among 187 New Zealand elite athletes, 21% of participants reported symptoms consistent with depression, while only 5% of them were taking an antidepressant medication.[133] Stigma, poor mental health literacy, self-reliance, negative experience of help-seeking in the past are known barriers for seeking mental health assistance, whereas a positive past experience and a close relationship with a mental health professional were shown to be an enhancing factor for help-seeking among athletes.[134] Mental health service availability can be another constraint for athletes and coaches in many places.[135] However, competitive environment of athletes discouraging treatment seeking due to fear of being judged, embarrassed, bullied, or considered weak is counterbalanced with rising awareness of mental health problems and broadening service accessibility in recent years.[21,22] A longitudinal analysis (2011–2019) of US collegiate athletes showed increasing treatment-seeking rates (30%–35%) and future treatment consideration (65%–70%), although athletes reported lower rates of past treatment (32.7% vs. 42.0%) and consideration of future treatment in (69.5% vs. 74.6%) comparison to non-athletes.[22]

Family support can be critical in prevention strategies.[61] Coaches and other staff members who work with athletes play a critical role in reducing suicidality. Therefore, their involvement in limiting unnecessary stressors, enhancing social connectivity inside of the team, emotional support and problem acknowledgement may be beneficial in screening, prevention, and treatment adjunction.[131] Educating the professional team surrounding the athlete as well as the sports team about their fundamental role in the mental health of their players or teammates is therefore an important issue.[55]

Open conversations without stigmatization are crucial for the screening and prevention of suicide among professional athletes. Some experts advocated for screening for mental disorders in athletes as an integral part of

TABLE 3 Recommendations for suicide screening and prevention in elite athletes

Provide a detailed review of general and specific individual and environmental risk factors
Screen for TBI of any degree, frequency, and number
Screen for performance-enhancing drugs misuse (especially anabolic-androgenic steroids)
Inquire about life stressors and mental issues (especially in retired athletes)
Follow-up in retired athletes who had a mental or physical illness
Consider sport-related stressors: • playing position, • sports type, • phase of the competition, • career stage, • training load, • team environment, • physical or sexual abuse
Screen for substantial injuries, decreasing sports performance, or chronic pain
Encourage help-seeking for mental health issues • Limit stigma • Improve mental health service availability and accessibility as well as sports psychiatrists • Provide non-judgmental, open, and age-appropriate conversation • Involve coaches and staff members in limiting stressors and increasing awareness
Establish a working network between athletic teams, sports organizations, and medical centers

Abbreviation: TBI, traumatic brain injury.

practice with regular follow-ups.[18] The implementation of a systematic low-threshold psychiatric-psychotherapeutic care system for elite athletes should offer age-appropriate screening instruments, counseling, and early treatment that potentially can have a prophylactic effect on future mental health including suicidal ideation and behavior during or after the professional sports career. A detailed assessment for potential suicidal ideation and behavior should be included in an obligatory screening for mental health problems in professional and elite athletes along with the preparticipation physical examination. Athletic organizations, sports organizations, research institutes, and sports medicine clinics should work together to improve screening and intervention to diminish morbidity and mortality related to suicide (Table 3).

6 | PERSPECTIVE

The current stance of research indicates lower rates of suicides among elite athletes than in the general population. Nonetheless, considering the young age of most of the athletes along with individual and environmental risk factors specific to this group of the population, careful attention is required in assessing suicidality. Affective disorders, substance abuse, AAS misuse, TBI, sport-related injuries, life stressors, and retirement are the most prominent features of suicide risk among professional athletes, and hence should be the focus of screening tools. Future research should address existing gaps between suicide and CTE, as well as other presumed associations. Aside from designated screening, prevention strategies should also implement encouragement for help-seeking for mental health problems and illness, refinement of mental support services availability and accessibility, as well as sport psychiatrists and psychologists in the care teams, better cooperation with coaches and sports team members, as well as network establishment between medical and athletic organizations.

DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

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How to cite this article: Pichler E-M, Ewers S, Ajdacic-Gross V, et al. Athletes are not at greater risk for death by suicide: A review. *Scand J Med Sci Sports*. 2023;00:1-17. doi:10.1111/sms.14316